INDUSTRIAL INTERNSHIP REPORT

ON

QUALITY CONTROL AND ASSEMBLY OPERATIONS IN AUTOMOTIVE COMPONENT MANUFACTURING

An Industrial Internship Report

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ABSTRACT

This report presents the industrial internship training undergone at AUTOFIT Private Limited, Hosur Manufacturing Plant, from 01 June 2026 to 27 June 2026, as part of the academic curriculum of the III Year B.E. Mechanical Engineering programme at College of Engineering Guindy, Anna University, Chennai. AUTOFIT Private Limited is an automotive component manufacturer with facilities across India, producing two-wheeler seats, PU-moulded foam products, injection-moulded plastic components, wheel assemblies and handle assemblies. During the training, exposure was gained to the complete manufacturing flow, beginning with raw-material and incoming component inspection and proceeding through PU foam manufacturing, seat-cover cutting and stitching, seat assembly, wheel and handle assembly, injection moulding and welding operations. Considerable emphasis was placed on quality control, covering incoming, in-process and final inspection, visual and dimensional inspection methods, the use of vernier calipers and Go/No-Go gauges, torque control in fastening operations, and the procedure for defect identification, segregation and rework. The internship also provided insight into production-line balancing, 5S workplace organisation, industrial safety practices, material handling, maintenance of machinery, and the role of CAD-based engineering design. The training helped establish practical connections between subjects studied in the Mechanical Engineering curriculum — such as Manufacturing Technology, Metrology, Machine Design, Production Engineering and Industrial Safety — and their application on the shop floor. Overall, the internship bridged the gap between theoretical knowledge and industrial practice and provided a comprehensive understanding of how quality, cost and delivery are managed together in an automotive component manufacturing environment.

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1. INTRODUCTION

Industrial training is an important component of engineering education because it provides students with an opportunity to understand how theoretical concepts are implemented in an actual industrial environment. Classroom education provides fundamental knowledge about manufacturing processes, materials, machines, metrology, production systems and quality control, whereas industrial training helps students understand how these concepts are integrated into a real production environment.

As part of my academic curriculum, I underwent internship training at **AUTOFIT Private Limited, Hosur**, from 01 June 2026 to 27 June 2026. The internship was primarily focused on **quality control and assembly operations in automotive component manufacturing**. During the training period, I was exposed to various manufacturing and assembly activities and studied the methods followed to maintain product quality and production efficiency.

The internship provided practical exposure to the sequence of operations involved in automotive component manufacturing. I was able to understand how raw materials and individual components move through different stages of production before becoming finished products. Particular attention was given to assembly operations, inspection procedures, workplace safety, material handling and quality control practices.

The training also helped me understand the importance of consistency in mass production. In an automotive manufacturing environment, even a small dimensional or assembly error can affect the performance, appearance or compatibility of the final product. Therefore, inspection and process control are required at different stages rather than depending only on final inspection.

2. COMPANY PROFILE

AUTOFIT Private Limited is an automotive component manufacturing company involved in the development and production of various components and assemblies used in automobiles. The company's product portfolio includes two-wheeler seats, wheel assemblies, handle assemblies, PU-moulded foam components, injection-moulded plastic components and seating systems for off-road vehicles and other automotive applications.

The company operates multiple manufacturing facilities in India, including its manufacturing facility at Hosur, Tamil Nadu. The Hosur plant provides an industrial environment where different manufacturing, assembly and quality-control activities are carried out to meet automotive production requirements.

AUTOFIT's product range demonstrates the integration of several manufacturing technologies within the automotive component industry. Depending on the product, operations may involve metal fabrication, welding, plastic injection moulding, PU foam moulding, leather or PVC cutting, stitching, assembly and inspection. This makes the company a suitable environment for understanding the relationship between different manufacturing processes.

During my internship, I gained an overall understanding of how an automotive component manufacturing facility operates. Rather than considering each manufacturing process as an isolated activity, I studied how material flow, production, assembly and inspection are connected to form a continuous manufacturing system.

3. OBJECTIVES OF THE INTERNSHIP

The primary objective of the internship was to obtain practical exposure to automotive component manufacturing and understand the working principles of assembly and quality-control operations. The training was intended to bridge the gap between the theoretical knowledge acquired during Mechanical Engineering studies and its practical application in an industrial environment.

Another important objective was to understand the importance of quality at different stages of manufacturing. I learned that quality is not simply checked after a component has been completely manufactured. Instead, various checks and controls are incorporated throughout the production process to identify problems at an early stage.

The internship also aimed to provide exposure to industrial safety, production discipline, material handling and workplace organization. Studying actual production activities helped me understand the importance of standard procedures, proper tool usage, operator responsibility and communication between different sections.

4. VISIT TO THE MANUFACTURING PLANT

At the beginning of the internship, I was introduced to the organization, its manufacturing activities and the general safety requirements of the plant. The induction helped me understand the importance of following workplace rules and maintaining discipline within the production environment.

A general plant visit was conducted to familiarize me with the different sections. I was shown areas associated with production, assembly, inspection, material storage and finished-product handling. The movement of materials between different sections helped me understand the basic production flow.

One of the major observations during the plant visit was the structured arrangement of workstations. Each workstation was responsible for a particular operation, and the output from one stage was transferred to the following stage. This arrangement helps maintain continuous production and reduces unnecessary movement of components.

The plant visit also provided an opportunity to understand the difference between a classroom description of manufacturing and actual industrial practice. Factors such as production targets, operator efficiency, material availability, quality requirements and workplace safety all have to be considered simultaneously.

5. RAW MATERIAL HANDLING AND INSPECTION

Raw materials and bought-out components form the starting point of the manufacturing process. Before materials are introduced into production, their condition and conformity with the required specifications are important considerations. Poor-quality incoming material can result in defects later in the production process.

During the internship, I was introduced to the general procedure involved in handling incoming materials and components. Materials are identified, stored appropriately and supplied to the relevant production areas. Proper identification is important because automotive manufacturing often involves several similar-looking components having different specifications.

Inspection at the incoming stage helps prevent unsuitable materials from entering the production line. Depending on the material and component, inspection may involve checking dimensions, appearance, quantity, physical condition and other specified characteristics.

I understood that controlling the quality of raw materials is an important part of overall quality management. If a defective material reaches the production line, it may result in wasted labour, additional processing, rework or rejection of the finished product.

6. MANUFACTURING PROCESS OF AUTOMOTIVE COMPONENTS

Automotive component manufacturing involves a sequence of operations rather than a single manufacturing process. The exact sequence depends on the type of component being produced. Different components may require cutting, forming, moulding, welding, stitching, machining, fastening or assembly.

In the case of seating products, for example, different materials and components have to be prepared before they are brought together. The seat frame, foam cushion and covering material have separate manufacturing or preparation processes before final assembly.

The manufacturing process therefore requires coordination between different sections. If one section experiences a delay, the following section may also be affected. Similarly, if an incorrectly manufactured component reaches the assembly section, it can result in assembly problems or final-product rejection.

Through the internship, I understood the importance of maintaining a continuous and controlled material flow. Production planning, inspection and assembly operations have to work together to achieve the required production quantity while maintaining consistent quality.

7. TWO-WHEELER SEAT MANUFACTURING

Two-wheeler seats are one of the important product categories associated with AUTOFIT. The manufacture of a seat involves several stages because the final product is made from multiple components such as the frame, foam cushion and seat covering.

The general manufacturing sequence begins with the preparation of the seat frame and foam component. The foam provides cushioning and contributes significantly to the comfort of the rider. The foam must have the required shape and dimensions so that it can be properly fitted to the seat structure.

The covering material is separately prepared according to the required shape. Cutting accuracy is important because the cover must fit properly over the foam and frame. After cutting, the required pieces may be stitched together to form the complete seat cover.

During final assembly, the different components are brought together. The cover is positioned over the foam and frame and fixed using the appropriate assembly method. The completed seat is then inspected for appearance, alignment, assembly condition and other applicable requirements.

This process helped me understand that a finished automotive product is the result of several interconnected manufacturing operations. The quality of the final seat therefore depends on the quality and accuracy of each preceding process.

8. PU FOAM MANUFACTURING

Polyurethane foam is widely used in automotive seating because of its cushioning properties. It provides the required softness and support while also allowing the seat to retain its designed shape.

The general PU foam manufacturing process involves placing the mould in the required condition and introducing the appropriate PU material into the mould. The material expands and fills the mould cavity, after which it undergoes curing. Once the material has sufficiently cured, the mould is opened and the foam component is removed.

The shape of the mould determines the geometry of the resulting foam component. Therefore, proper mould condition and process control are important for obtaining consistent components. Variations in the process can potentially result in differences in shape, surface condition or other characteristics.

During the internship, I gained an understanding of the importance of moulding processes in automotive component manufacturing. I also learned that inspection is required after moulding to ensure that the component is suitable for the next manufacturing stage.

9. SEAT COVER CUTTING AND STITCHING

The seat covering is an important part of the final product because it contributes to both appearance and durability. The material must be cut according to the required pattern before it can be assembled around the foam and frame.

Cutting requires proper positioning of the material and accurate following of the required profile. An incorrectly cut piece can create problems during stitching or final assembly. Therefore, the dimensions and condition of the cut pieces need to be controlled.

After cutting, individual pieces are joined through stitching. Proper alignment of the pieces is important to achieve the required appearance. The stitching must also be sufficiently consistent to provide an acceptable and durable joint.

During inspection of stitched components, aspects such as alignment, stitching quality, loose threads, surface appearance and other visible defects can be considered. This showed me how apparently simple operations such as cutting and stitching can have a significant effect on the quality of the finished automotive component.

10. SEAT ASSEMBLY OPERATIONS

Assembly is the stage at which individual components are combined to produce the completed product. In seat manufacturing, components such as the frame, foam and cover have to be correctly positioned and secured.

The assembly operation requires proper sequencing because the components need to be installed in the correct order. Incorrect positioning can result in poor appearance, improper fit or functional problems. Therefore, operators follow established procedures to maintain consistency.

I understood the importance of operator skill in assembly operations. Even when components are manufactured accurately, improper assembly can introduce defects into the final product. This is why standardized work instructions and inspection procedures are important in mass production.

The completed assembly is subjected to inspection before it is transferred to the next stage. This helps identify defects before the product reaches packaging or dispatch.

11. WHEEL AND HANDLE ASSEMBLY

Apart from seating products, AUTOFIT is also involved in manufacturing automotive assemblies such as two-wheeler wheel assemblies and handle assemblies. These products consist of several individual components that need to be correctly positioned and fastened.

In an assembly operation, component identification is important because similar-looking parts may have different dimensions or applications. Correct positioning and fastening are also necessary to ensure that the completed assembly satisfies the required specifications.

The assembly process demonstrated the importance of standardization in mass production. Performing the same sequence of operations consistently helps reduce variation between products.

Quality checks are performed to identify issues such as incorrect positioning, missing components, improper fastening, dimensional problems or visible defects. This reinforced the importance of inspection as an integral part of assembly manufacturing.

12. INJECTION MOULDING

Injection moulding is a manufacturing process used to produce plastic components with a defined geometry. The process begins with plastic material being heated until it reaches a suitable molten state. The molten material is then injected into a mould cavity under pressure.

After the material fills the mould, it is allowed to cool and solidify. The mould is then opened and the finished component is removed. The component may subsequently undergo trimming, inspection or further assembly depending on its application.

The advantage of injection moulding is that large quantities of similar components can be produced with consistent geometry when the process is properly controlled. However, process parameters, mould condition and material properties can influence the quality of the final component.

Studying this process helped me understand how polymer materials studied in engineering can be processed into practical automotive components.

13. WELDING OPERATIONS

Welding is used to permanently join metallic components and is an important process in the production of various automotive structures and assemblies. During the internship, I was introduced to the role of welding in automotive component manufacturing.

The quality of a welded joint depends on several factors, including the condition of the materials, welding parameters, joint preparation and operator technique. An improperly produced weld can affect the strength and reliability of the component.

Visual inspection is commonly used to identify obvious weld-related defects such as excessive spatter, irregular weld appearance or visible discontinuities. Depending on the application, additional inspection methods may also be used.

Learning about welding operations helped me relate manufacturing processes studied during Mechanical Engineering with their application in automotive component production.

14. QUALITY CONTROL

Quality control was one of the most important areas of my internship. In automotive manufacturing, quality control ensures that components and finished products meet the required specifications before they proceed to the next stage or reach the customer.

Quality control can take place at different stages. Incoming inspection checks materials and components received from suppliers, in-process inspection checks products during manufacturing, and final inspection verifies the completed product.

I learned that quality control is not simply a method for rejecting defective products. Its larger purpose is to identify sources of variation and prevent defects from continuing through the production process.

This approach is particularly important in mass production because a small process problem can affect a large number of components if it is not identified early.

15. VISUAL INSPECTION

Visual inspection is one of the basic methods used for identifying defects in manufactured components. It can be performed at different stages of production and is especially useful for identifying visible surface and assembly problems.

During inspection, the component can be checked for scratches, dents, surface marks, contamination, improper stitching, wrinkles, missing parts, incorrect assembly and other visible abnormalities.

Although visual inspection does not provide numerical dimensional information, it is highly useful for identifying appearance-related and assembly-related defects. It can also be performed quickly in a production environment.

I learned that visual inspection requires attention to detail and knowledge of the acceptable condition of the product. The inspector must be able to distinguish between acceptable variation and an actual defect.

16. DIMENSIONAL INSPECTION

Dimensional inspection is performed to determine whether a component has been manufactured within its specified dimensions and tolerances. Dimensional accuracy is particularly important in automotive manufacturing because individual components must fit correctly during assembly.

Instruments such as vernier calipers, micrometers, gauges and other measuring devices can be used depending on the required measurement. A vernier caliper, for example, can be used to measure external dimensions, internal dimensions and depth.

I was able to relate this activity to the concepts of **Metrology and Measurements** studied in Mechanical Engineering. The practical environment helped me understand why tolerances are necessary in manufacturing and why components cannot simply be manufactured to an ideal dimension without considering permissible variation.

Dimensional inspection also helps identify process-related problems. If measurements consistently move away from the required range, corrective action may be necessary in the corresponding manufacturing process.

17. GO AND NO-GO GAUGES

Go and No-Go gauges are used as a rapid method of checking whether a component satisfies a particular dimensional requirement. The principle is based on checking the component against predetermined limits rather than obtaining an exact numerical measurement.

The **GO** portion is designed to represent one limiting condition and should pass through or fit the relevant feature. The **NO-GO** portion represents the opposite limiting condition and should not pass.

This method is useful in mass production because inspection can be performed rapidly and consistently. It reduces the time required for measuring individual components when only conformity to a particular tolerance range needs to be established.

Studying this method gave me a better understanding of how theoretical concepts of limits, fits and tolerances are applied in practical manufacturing.

18. TORQUE CONTROL IN ASSEMBLY

Fasteners are frequently used in automotive assemblies, and their tightening condition is important for the reliability of the product. A fastener that is insufficiently tightened may loosen during operation, whereas excessive tightening may damage the fastener or the component.

Torque-controlled tools can be used to maintain a specified tightening condition. This provides better consistency than relying entirely on the operator's judgement.

During the internship, I understood the importance of controlling assembly parameters rather than focusing only on the final appearance of the product. Correct fastening contributes to the structural integrity and reliability of the completed assembly.

This was also a practical application of mechanical engineering concepts related to torque, friction, fasteners and joint integrity.

19. DEFECT IDENTIFICATION AND REWORK

Defect identification is an important part of quality control. When a component does not meet the applicable requirements, it needs to be identified and separated from accepted components to prevent accidental use.

Depending on the nature of the defect, a component may be suitable for rework or may need to be rejected. Rework involves correcting a defect according to an approved procedure and subsequently inspecting the component again.

This process demonstrates the importance of traceability and segregation in manufacturing. Mixing rejected or defective components with acceptable products can create additional quality problems.

I learned that the objective of quality control is not merely to detect defects at the final stage, but to minimize the occurrence of defects by controlling the manufacturing process itself.

20. PRODUCTION LINE AND WORKSTATION MANAGEMENT

Production in an automotive component plant is generally divided into multiple workstations. Each workstation performs a specific operation before passing the component to the next stage.

This arrangement allows repetitive operations to be organized systematically. However, the time required at each workstation needs to be considered. If one operation takes significantly longer than the others, it can become a bottleneck and reduce the overall production rate.

This is related to the concept of **line balancing**, where work is distributed between stations to achieve a more uniform production flow. The concept of takt time is also useful for understanding the required production rate based on customer demand.

Studying the production line helped me understand how production engineering concepts are applied in an actual manufacturing environment.

21. 5S AND WORKPLACE ORGANIZATION

Workplace organization plays an important role in maintaining productivity and safety. During the internship, I was introduced to the principles of **5S**, which are commonly associated with systematic workplace organization.

The five principles are **Sort, Set in Order, Shine, Standardize and Sustain**. The objective is to remove unnecessary items, arrange required items systematically, maintain cleanliness, establish standard practices and ensure that these practices are continuously followed.

A well-organized workstation reduces unnecessary movement and makes tools and materials easier to locate. It can also reduce the possibility of material mix-ups and improve workplace safety.

The industrial environment helped me understand that 5S is not simply about keeping a workplace clean. It is a systematic approach to improving efficiency, organization and discipline.

22. INDUSTRIAL SAFETY

Safety is a fundamental requirement in a manufacturing environment because workers may be exposed to moving machinery, tools, hot surfaces, electrical equipment, chemicals and material-handling activities.

During the internship, I was introduced to general industrial safety practices and the importance of following the instructions provided by the organization. Appropriate personal protective equipment must be used wherever required, and workers must remain within designated safe areas.

Machine guarding and safe operating procedures are particularly important around production equipment. Unauthorized operation of machinery should be avoided, and workers must understand the potential hazards associated with their work area.

The internship helped me understand that industrial productivity cannot be separated from worker safety. A safe production environment reduces accidents, equipment damage and production interruptions.

23. MATERIAL HANDLING AND STORAGE

Material handling is necessary throughout the manufacturing process because raw materials, work-in-progress components and finished products need to move between different sections.

Materials may be stored using racks, bins, pallets and designated storage areas. Appropriate identification and arrangement of materials help prevent component mix-ups and unnecessary movement.

Efficient material handling can also reduce production delays. If the required component is not available at a workstation at the appropriate time, the production process can be interrupted even when the machines and operators are ready.

Through this study, I understood that manufacturing efficiency depends not only on machines and production processes but also on proper logistics and material management.

24. MAINTENANCE OF MACHINERY

Manufacturing equipment requires regular maintenance to ensure reliable operation. Unexpected machine breakdowns can affect production targets, increase downtime and potentially affect delivery schedules.

Maintenance activities can generally be classified into preventive and corrective maintenance. Preventive maintenance is carried out at planned intervals to reduce the likelihood of failure, while corrective maintenance is performed when a fault or failure occurs.

During the internship, I understood the importance of maintaining production equipment in suitable operating condition. Machine reliability is directly connected to production continuity and product quality.

This also demonstrated the practical importance of mechanical engineering knowledge related to machines, mechanisms, lubrication, wear and maintenance.

25. ROLE OF CAD AND ENGINEERING DESIGN

Automotive component manufacturing begins long before the physical production process. The component must first be designed according to customer requirements and engineering specifications.

CAD software is used extensively in modern product development to create and modify component designs. As per AUTOFIT's published information, its engineering activities include CAD-based product development, customer-specific requirements, benchmarking and value engineering.

The connection between design and manufacturing was an important aspect of my learning. A component cannot be considered only from the viewpoint of its shape. The design must also consider manufacturing feasibility, assembly requirements, material selection, dimensional tolerances and quality requirements.

This helped me understand how Mechanical Engineering subjects such as engineering drawing, CAD, manufacturing technology and machine design are interconnected in an industrial environment.

26. QUALITY, COST AND DELIVERY

One of the important lessons obtained from the internship was the relationship between **quality, cost and delivery**.

A manufacturing organization has to produce the required quantity within the required time while maintaining acceptable quality. Producing a large quantity of defective components does not represent successful production because defects result in rework, rejection, material wastage and additional labour.

Similarly, excessive inspection or inefficient processes can increase manufacturing costs. Therefore, the objective is to develop controlled processes that consistently produce acceptable products.

The internship helped me understand that quality has to be built into the manufacturing process rather than depending entirely on final inspection.

27. COMMON MANUFACTURING DEFECTS

During the study of manufacturing and assembly operations, I learned about several types of defects that can occur in automotive components. These include dimensional variations, surface defects, incorrect assembly, improper fastening, stitching defects, material damage and moulding-related defects.

For seating components, visible defects may include wrinkles, improper stitching, surface damage or incorrect cover positioning. For metallic components, defects may include dimensional variation, scratches or welding-related problems. Plastic components may experience issues related to moulding, surface condition or dimensional accuracy.

The important point is that each defect can have a different cause. Therefore, identifying the defect is only the first step. Understanding the cause of the defect is necessary for preventing it from occurring repeatedly.

28. INTERNSHIP LEARNING EXPERIENCE

The internship provided me with practical exposure to the working environment of an automotive component manufacturing organization. I learned that manufacturing involves much more than operating machines. It requires coordination between production, quality, maintenance, material handling, engineering and other departments.

I gained a better understanding of how individual components move through different stages before becoming a finished product. Studying assembly and inspection activities helped me understand why manufacturing specifications and tolerances are important.

The internship also improved my awareness of workplace discipline. In an industrial environment, punctuality, safety, proper communication and adherence to standard procedures are important for maintaining smooth operations.

29. APPLICATION OF MECHANICAL ENGINEERING KNOWLEDGE

The internship allowed me to relate several subjects from the Mechanical Engineering curriculum to industrial applications.

Concepts from **Manufacturing Technology** were studied through moulding, welding, cutting and assembly operations. **Metrology and Measurements** were related to dimensional inspection and the use of measuring instruments.

Concepts from **Production Engineering** were applicable to production flow, workstation organization, line balancing and productivity. **Engineering Materials** were relevant to the use of steel, plastics, PU foam and covering materials.

The internship also demonstrated the practical connection between **CAD, machine design, quality control and manufacturing**. This integrated understanding would be difficult to obtain through theoretical study alone.

30. SKILLS ACQUIRED DURING THE INTERNSHIP

The internship improved my understanding of automotive manufacturing and gave me practical exposure to industrial quality-control practices. I developed a better understanding of how manufacturing processes are organized and how individual operations contribute to the final product.

I also gained familiarity with industrial terminology such as **assembly line, inspection, tolerance, rework, rejection, quality control, 5S, line balancing, production flow and preventive maintenance**.

Apart from technical knowledge, I gained experience in observing workplace procedures, communicating with personnel, following safety requirements and understanding the importance of teamwork in an industrial organization.

31. OVERALL OBSERVATIONS

One of my major observations during the internship was that consistency is extremely important in automotive manufacturing. When thousands of similar components are manufactured, each component needs to satisfy the same basic requirements.

I also observed that quality and production are closely related. Increasing production speed without maintaining process control can increase defects, while excessive inspection without efficient production methods can increase cost and delay delivery.

Another important observation was the importance of human involvement. Even with automated and semi-automated equipment, operators, inspectors, maintenance personnel and engineers play important roles in maintaining production efficiency and quality.

32. CONCLUSION

The industrial internship at **AUTOFIT Private Limited, Hosur** provided me with valuable practical exposure to automotive component manufacturing. The internship focused primarily on **quality control and assembly operations**, while also providing an understanding of manufacturing processes, material handling, production flow, safety and workplace organization.

During the training period, I gained exposure to the manufacturing and assembly of automotive components and developed an understanding of processes such as PU foam manufacturing, seat-cover preparation, stitching, assembly, injection moulding and welding. I also learned about visual inspection, dimensional inspection, defect identification, rework and quality-control procedures.

The internship helped me understand that manufacturing is an interconnected system in which design, materials, production, assembly, inspection, maintenance and logistics all contribute to the final product. A failure or variation at any stage can influence subsequent operations and ultimately affect product quality.

The experience also allowed me to connect my academic knowledge in Mechanical Engineering with actual industrial practices. Concepts related to manufacturing technology, metrology, materials, production engineering, CAD, machine design and industrial safety became more understandable through practical observation.

Overall, the internship was a valuable learning experience that improved my technical awareness and provided a better understanding of the working environment in the automotive component manufacturing industry.

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